

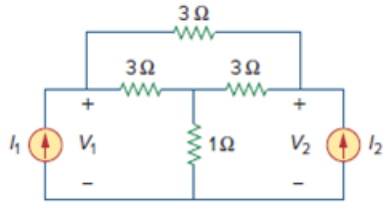
Problem

In the bridge circuit of Fig. 19.87, $I_1 = 20$ A and $I_2 = -8$ A.

(a) Find V_1 and V_2 using y parameters.

(b) Confirm the results in part (a) by direct circuit analysis.

Figure 19.87



Step-by-step solution

Step 1 of 18

Refer to Figure 19.87 in the text book.

Consider the circuit diagram shown in Figure 1.

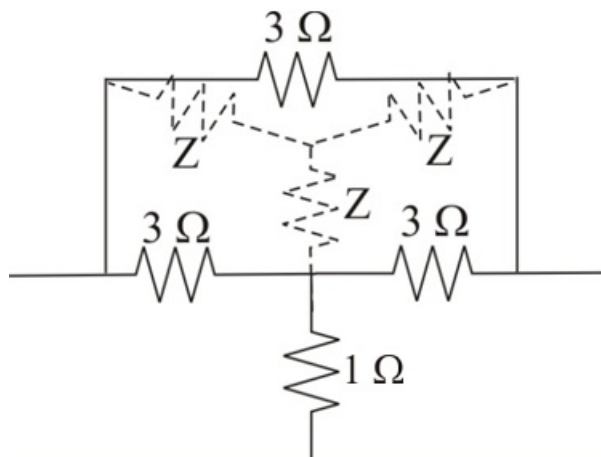


Figure 1

Step 2 of 18

Convert the delta connected resistors to wye connection. All the values of the resistances in wye connection are equal since values of the resistances in delta connection are equal.

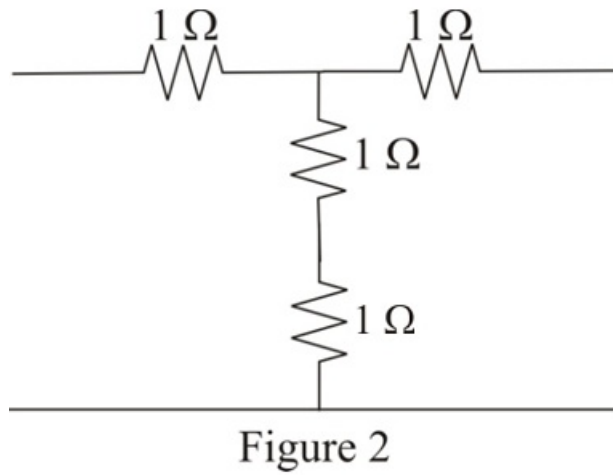
Calculate the value of each resistance in wye connection.

$$\begin{aligned} Z &= \frac{(3)(3)}{3+3+3} \\ &= \frac{9}{9} \\ &= 1 \, \Omega \end{aligned}$$

Therefore, the value of each resistance in wye connection is $1 \, \Omega$.

Step 3 of 18

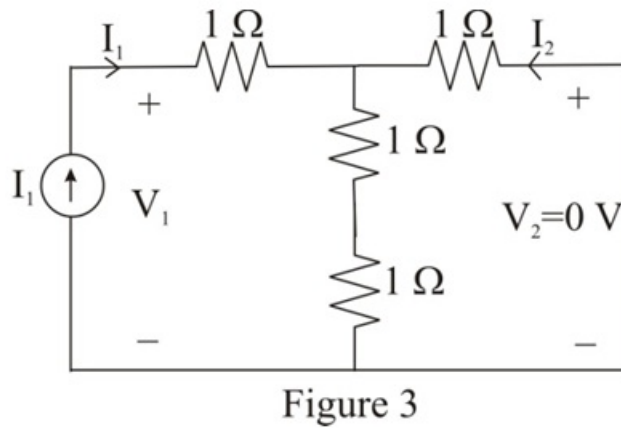
Redraw the circuit diagram as shown in Figure 2.



Step 4 of 18

(a)

Connect the current source I_1 as shown in Figure 1. Identify the currents and voltages in the circuit.



Step 5 of 18

Consider the expression for the voltage V_1 .

$$\begin{aligned} V_1 &= I_1 (1 + [1 \parallel (1+1)]) \\ &= I_1 (1 + [1 \parallel 2]) \\ &= I_1 \left(1 + \frac{(1)(2)}{1+2} \right) \\ &= I_1 \left(1 + \frac{2}{3} \right) \end{aligned}$$

$$\begin{aligned} V_1 &= I_1 (1 + 0.667) \\ &= I_1 (1.667) \end{aligned}$$

Therefore, the expression for the voltage V_1 is $I_1 (1.667)$.

Step 6 of 18

Consider the expression for the current I_2 .

$$\begin{aligned} I_2 &= -I_1 \left(\frac{1+1}{1+1+1} \right) \\ &= -I_1 \left(\frac{2}{3} \right) \\ &= -I_1 (0.667) \end{aligned}$$

Therefore, the expression for the current I_2 is $-I_1 (0.667)$.

Step 7 of 18

Determine the expression for the short circuit input admittance y_{11} .

$$y_{11} = \frac{I_1}{V_1}$$

Substitute $I_1 (1.667)$ for V_1 .

$$\begin{aligned} Y_{11} &= \frac{I_1}{I_1 (1.667)} \\ &= 600 \text{ mS} \end{aligned}$$

Therefore, the expression for the short circuit input admittance y_{11} is, 600 mS.

Step 8 of 18

Determine the value of the short circuit transfer admittance y_{21} from port-2 to port-1.

$$y_{21} = \frac{I_2}{V_1}$$

Substitute $I_1 (1.667)$ for V_1 and $-I_1 (0.667)$ for I_2

$$\begin{aligned} y_{21} &= \frac{-I_1 (0.667)}{I_1 (1.667)} \\ &= -400 \text{ mS} \end{aligned}$$

Therefore, the value of the short circuit transfer admittance y_{21} from port-2 to port-1 is, -400 mS.

Step 9 of 18

Connect the current source I_2 as shown in Figure 2. Identify the currents and voltages in the circuit.

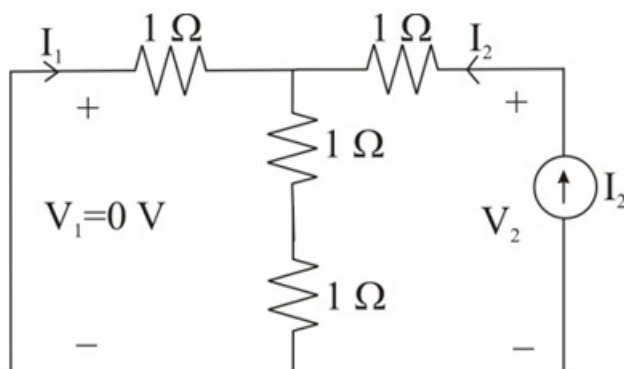


Figure 4

Consider the expression for the voltage V_2 .

$$\begin{aligned} V_2 &= I_2 \left(1 + [1 \parallel (1+1)] \right) \\ &= I_2 \left(1 + [1 \parallel 2] \right) \\ &= I_2 \left(1 + \frac{(1)(2)}{1+2} \right) \\ &= I_2 \left(1 + \frac{2}{3} \right) \end{aligned}$$

$$\begin{aligned} V_2 &= I_2 (1 + 0.667) \\ &= I_2 (1.667) \end{aligned}$$

Therefore, the expression for the voltage V_2 is $I_2 (1.667)$.

Step 11 of 18

Consider the expression for the current I_1 .

$$\begin{aligned} I_1 &= -I_2 \left(\frac{1+1}{1+1+1} \right) \\ &= -I_2 \left(\frac{2}{3} \right) \\ &= -I_2 (0.667) \end{aligned}$$

Therefore, the expression for the current I_1 is $-I_2 (0.667)$.

Step 12 of 18

Determine the value of the short circuit transfer admittance y_{12} from port-1 to port-2.

$$y_{12} = \frac{I_1}{V_2}$$

Substitute $I_2 (1.667)$ for V_2 and $-I_2 (0.667)$ for I_1 .

$$\begin{aligned} y_{12} &= -\frac{I_2 (0.667)}{I_2 (1.667)} \\ &= -400 \text{ mS} \end{aligned}$$

Therefore, the value of the short circuit transfer admittance y_{12} from port-1 to port-2 is, -400 mS .

Step 13 of 18

Determine the expression for the short circuit output admittance y_{22} .

$$y_{22} = \frac{I_2}{V_2}$$

Substitute $I_2 (1.667)$ for V_2 .

$$\begin{aligned} y_{22} &= \frac{I_2}{I_2 (1.667)} \\ &= 600 \text{ mS} \end{aligned}$$

Therefore, the expression for the short circuit output admittance y_{22} is, 600 mS .

Consider the equations of the admittance circuit.

$$\mathbf{I}_1 = \mathbf{y}_{11} \mathbf{V}_1 + \mathbf{y}_{12} \mathbf{V}_2$$

$$\mathbf{I}_2 = \mathbf{y}_{21} \mathbf{V}_1 + \mathbf{y}_{22} \mathbf{V}_2$$

Represent the equations in matrix form.

$$\begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{y}_{11} & \mathbf{y}_{12} \\ \mathbf{y}_{21} & \mathbf{y}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{y}_{11} & \mathbf{y}_{12} \\ \mathbf{y}_{21} & \mathbf{y}_{22} \end{bmatrix}^{-1} \cdot \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \frac{1}{(\mathbf{y}_{22}\mathbf{y}_{11} - (-\mathbf{y}_{21})(-\mathbf{y}_{12}))} \begin{bmatrix} \mathbf{y}_{22} & -\mathbf{y}_{12} \\ -\mathbf{y}_{21} & \mathbf{y}_{11} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \frac{1}{(\mathbf{y}_{22}\mathbf{y}_{11} - \mathbf{y}_{12}\mathbf{y}_{21})} \begin{bmatrix} \mathbf{y}_{22} & -\mathbf{y}_{12} \\ -\mathbf{y}_{21} & \mathbf{y}_{11} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix} \dots\dots (1)$$

Step 15 of 18

Recall equation (1).

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \frac{1}{(\mathbf{y}_{22}\mathbf{y}_{11} - \mathbf{y}_{12}\mathbf{y}_{21})} \begin{bmatrix} \mathbf{y}_{22} & -\mathbf{y}_{12} \\ -\mathbf{y}_{21} & \mathbf{y}_{11} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \end{bmatrix}$$

Substitute 600 mS for $\mathbf{y}_{22}$ and $\mathbf{y}_{11}$, -400 mS for $\mathbf{y}_{21}$ and $\mathbf{y}_{12}$, 20 A for $\mathbf{I}_1$ and -8 A for $\mathbf{I}_2$.

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \frac{1}{\begin{pmatrix} (600 \text{ m}) & -(-400 \text{ m}) \\ (-400 \text{ m}) & (600 \text{ m}) \end{pmatrix}} \begin{bmatrix} (600 \text{ m}) & -(-400 \text{ m}) \\ -(-400 \text{ m}) & (600 \text{ m}) \end{bmatrix} \cdot \begin{bmatrix} 20 \\ -8 \end{bmatrix}$$

$$= \frac{1}{(0.36 - 0.16)} \cdot \begin{bmatrix} 20(600 \text{ m}) - 8(400 \text{ m}) \\ 20(400 \text{ m}) - 8(600 \text{ m}) \end{bmatrix}$$

$$= \frac{1}{0.2} \cdot \begin{bmatrix} 12 - 3.2 \\ 8 - 4.8 \end{bmatrix}$$

$$= \frac{1}{0.2} \cdot \begin{bmatrix} 8.8 \\ 3.2 \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} 44 \\ 16 \end{bmatrix}$$

Therefore, the values of $\mathbf{V}_1$ and $\mathbf{V}_2$ are **44 V** and **16 V** respectively.

Step 16 of 18

(b)

Consider the circuit diagram shown in Figure 5.

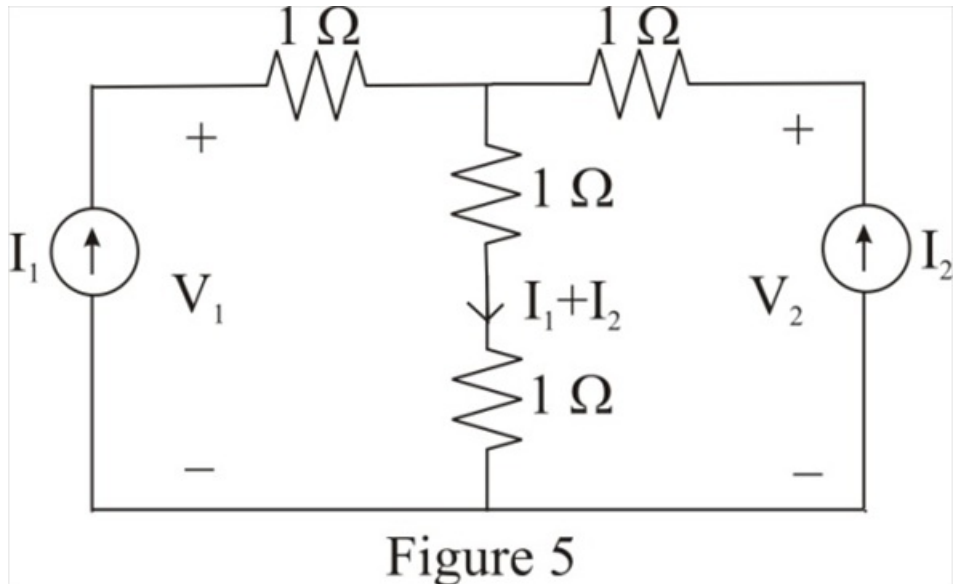


Figure 5

Step 17 of 18

Calculate the value of the voltage V_1 .

$$\begin{aligned} V_1 &= I_1(1) + (I_1 + I_2)(1+1) \\ &= I_1(1) + (I_1 + I_2)(2) \\ &= I_1(3) + I_2(2) \end{aligned}$$

Substitute 20 A for I_1 and -8 A for I_2 .

$$\begin{aligned} V_1 &= (20)(3) + (-8)(2) \\ &= 60 - 16 \\ &= 44 \text{ V} \end{aligned}$$

Therefore, the value of the voltage V_1 is **44 V**.

Step 18 of 18

Calculate the value of the voltage V_2 .

$$\begin{aligned} V_2 &= I_2(1) + (I_1 + I_2)(1+1) \\ &= I_2(1) + (I_1 + I_2)(2) \\ &= I_2(3) + I_1(2) \end{aligned}$$

Substitute 20 A for I_1 and -8 A for I_2 .

$$\begin{aligned} V_2 &= (-8)(3) + (20)(2) \\ &= -24 + 40 \\ &= 16 \text{ V} \end{aligned}$$

Therefore, the value of the voltage V_2 is **16 V**.